



Atomic Force Microscopy

Eka Nurfani

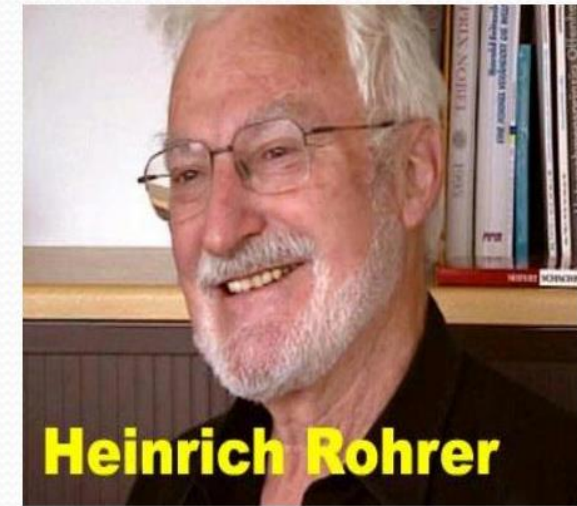
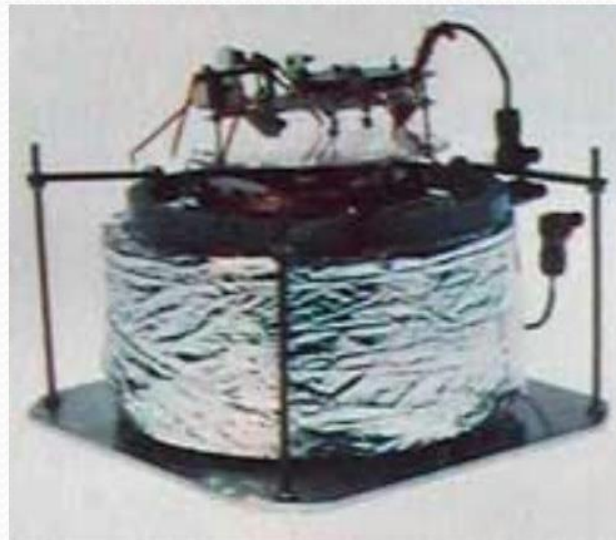
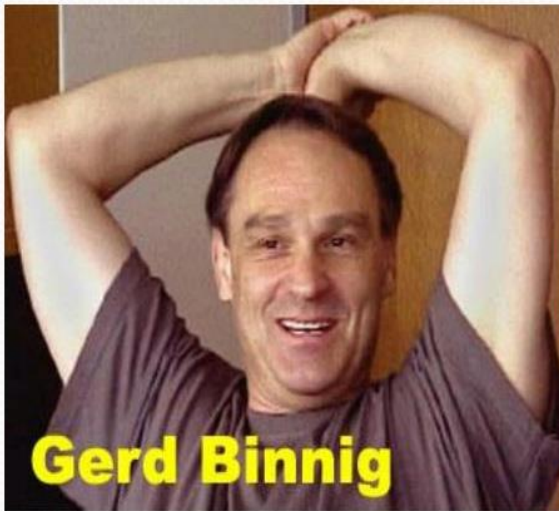
Disampaikan dalam Mata Kuliah Karakterisasi Material

Outline

- **What is AFM?**
 - Basic Principles
 - AFM Parts
- **Imaging modes**
 - Atomic Interaction (Van der Waals interaction)
 - Contact mode
 - Non-contact mode
 - Tapping mode
- **Summary**

History

- Scanning Probe Microscopy (SPM) was started with the invention of Scanning Tunneling Microscopy (STM) by Gert Binnig and Heinrich Rohrer in 1981.
- They received Nobel Prize in 1986 for the same.



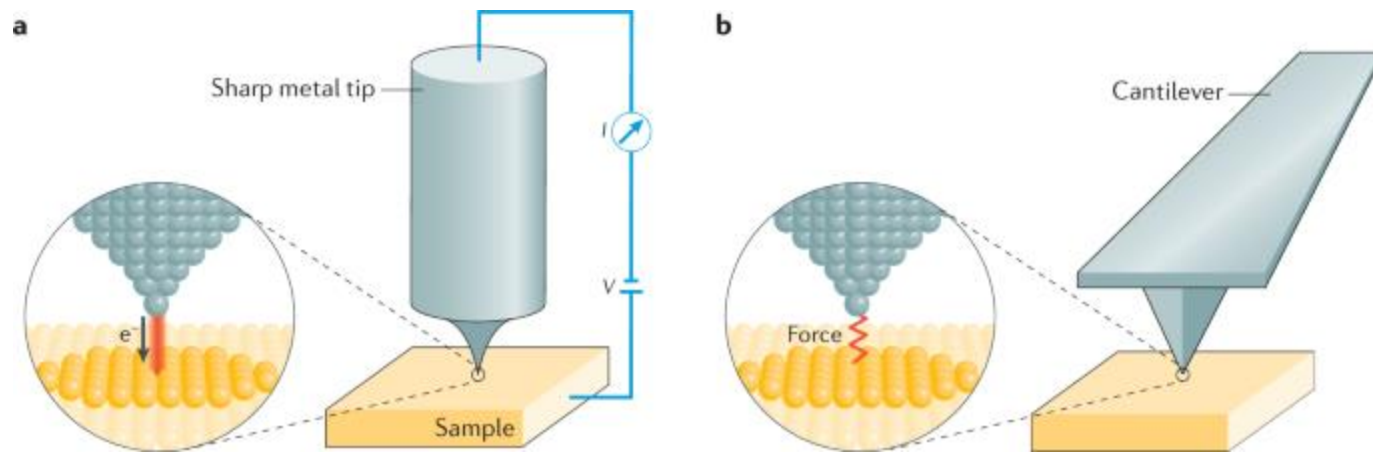
Binnig also invented the Atomic Force Microscope (AFM) at Stanford University in 1986.

<https://www.studypool.com>

Scanning Probe Microscopy (SPM)

The SPM system consists of:

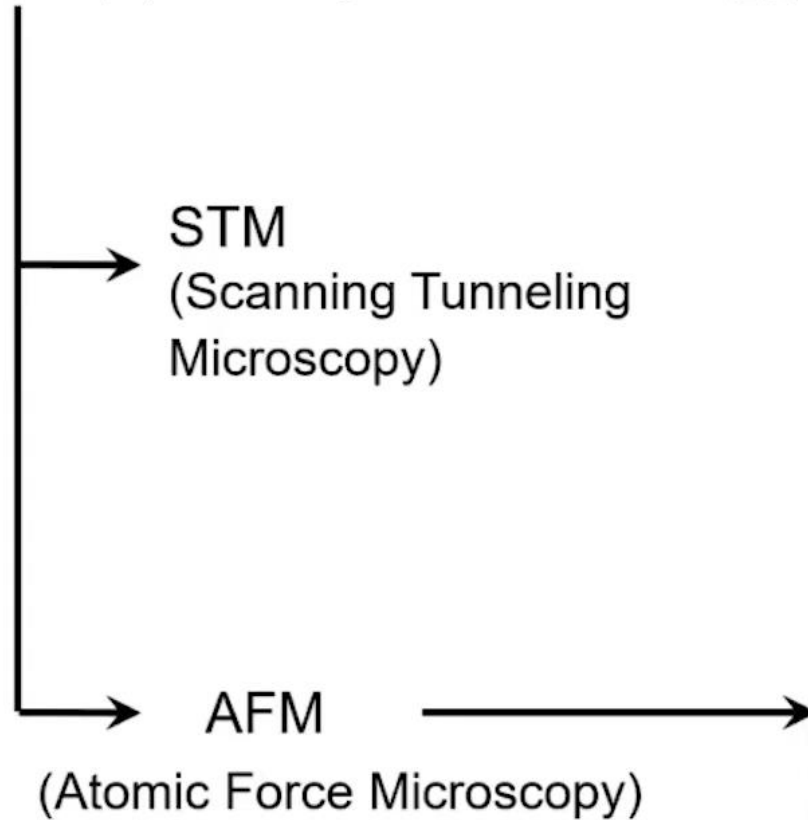
- **Probe** that are nanosized
- **Scanning and Feedback mechanism** → accurate to the subnanometer level (achieved with piezoelectric materials)
- **Highly sophisticated computer controls** (fast Digital to Analog converter, etc)



Bian, K., Gerber, C., Heinrich, A.J. *et al.* Scanning probe microscopy. *Nat Rev Methods Primers* **1**, 36 (2021)

SPM Family

SPM ; (Scanning Probe Microscopy)



Optional Modes

Electrical Property

- Electrostatic Force Microscopy(EFM)
- Kelvin Probe Force Microscopy (KPFM)
- Conductive AFM(I-AFM)
- Scanning Spreading-Resistance Microscopy (SSRM)
- Scanning Capacitance Microscopy(SCM)

Mechanical Property

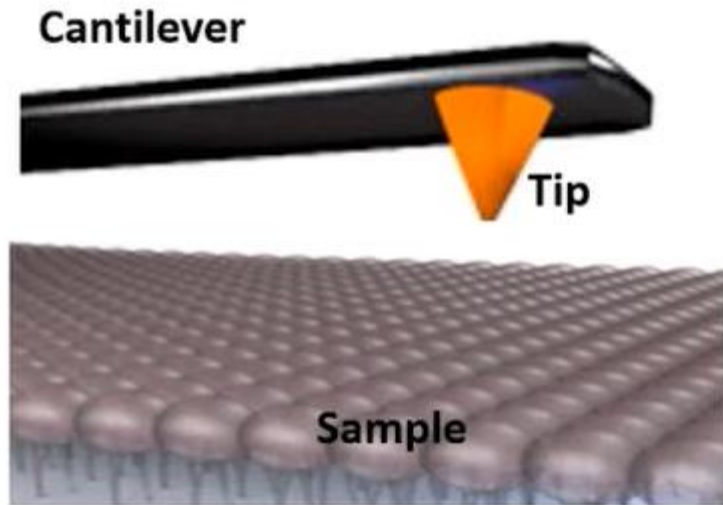
- Force spectroscopy
- PinPoint nanomechanical mode
- Lateral Force Microscopy (LFM)
- Lithography

Thermal Property

- Scanning Thermal Microscopy(SThM)

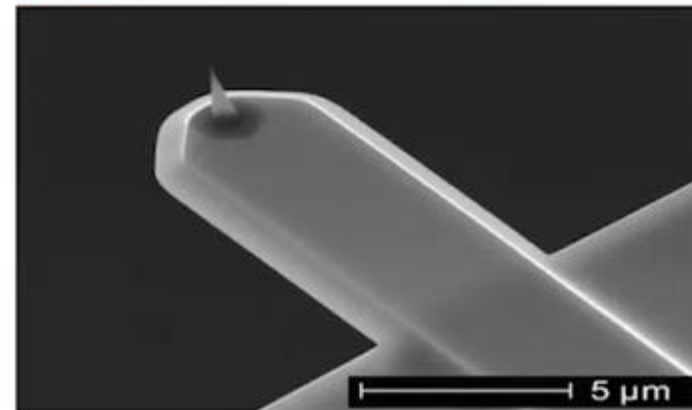
How does the AFM actually generate an image?

The AFM generates images by 'feeling' the surface of the sample.



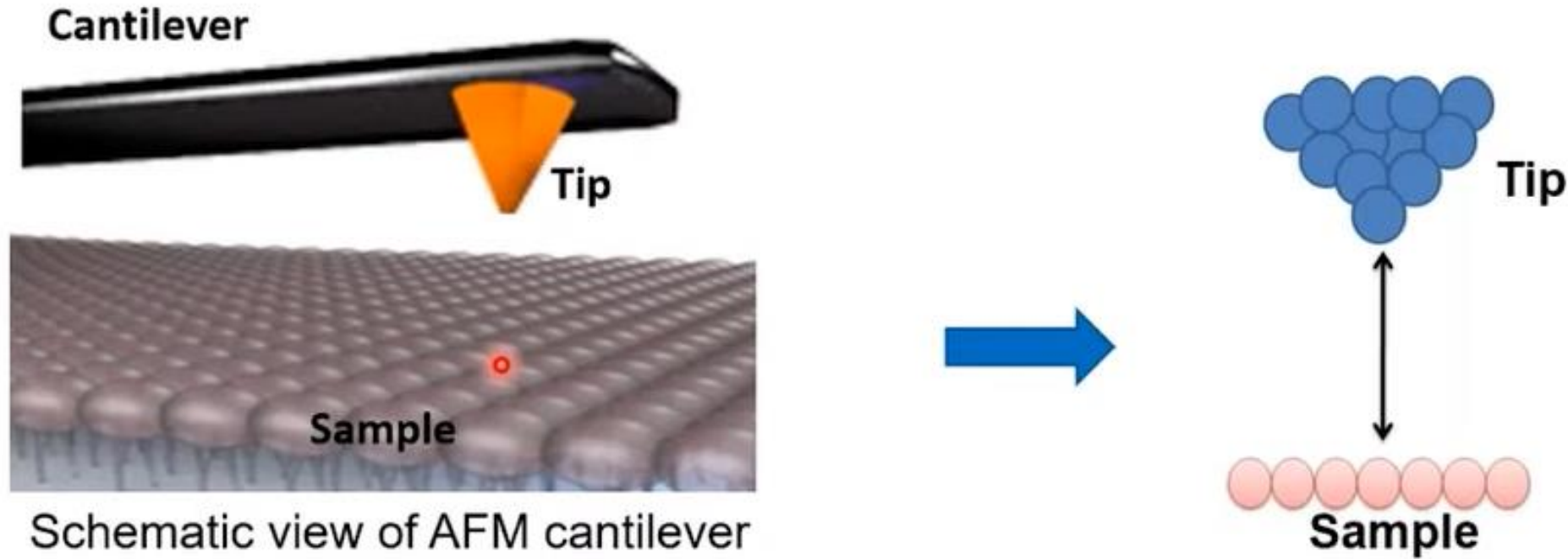
Schematic view of AFM cantilever

Atomic Force Microscopy (AFM) is not like typical optical microscopy, but it is like a **probing robot**.



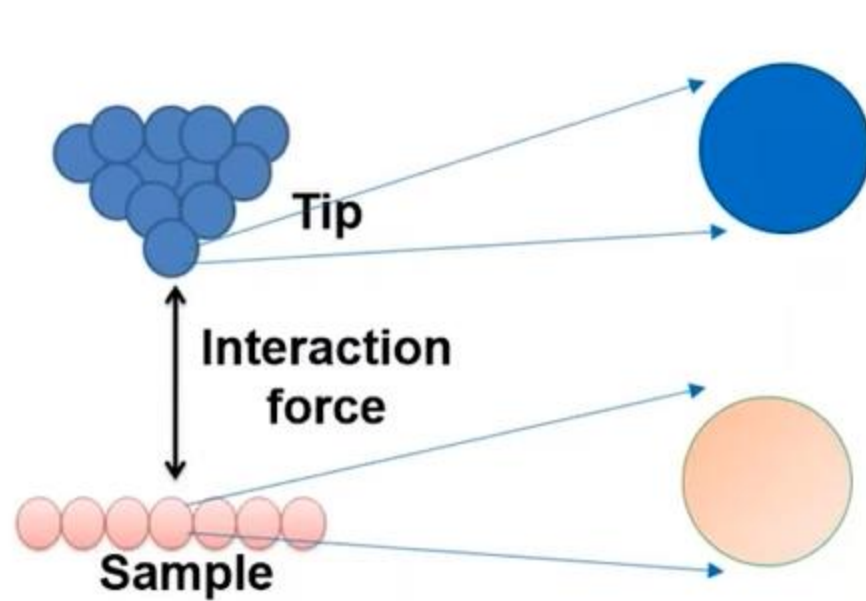
SEM image of AFM cantilever

What does the AFM tip “feel”?

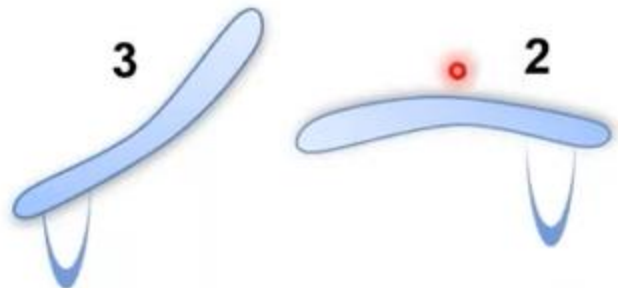


The name involves the word “force” in atomic force microscope which indicate that the interactions between the tip and the sample are due to some forces.

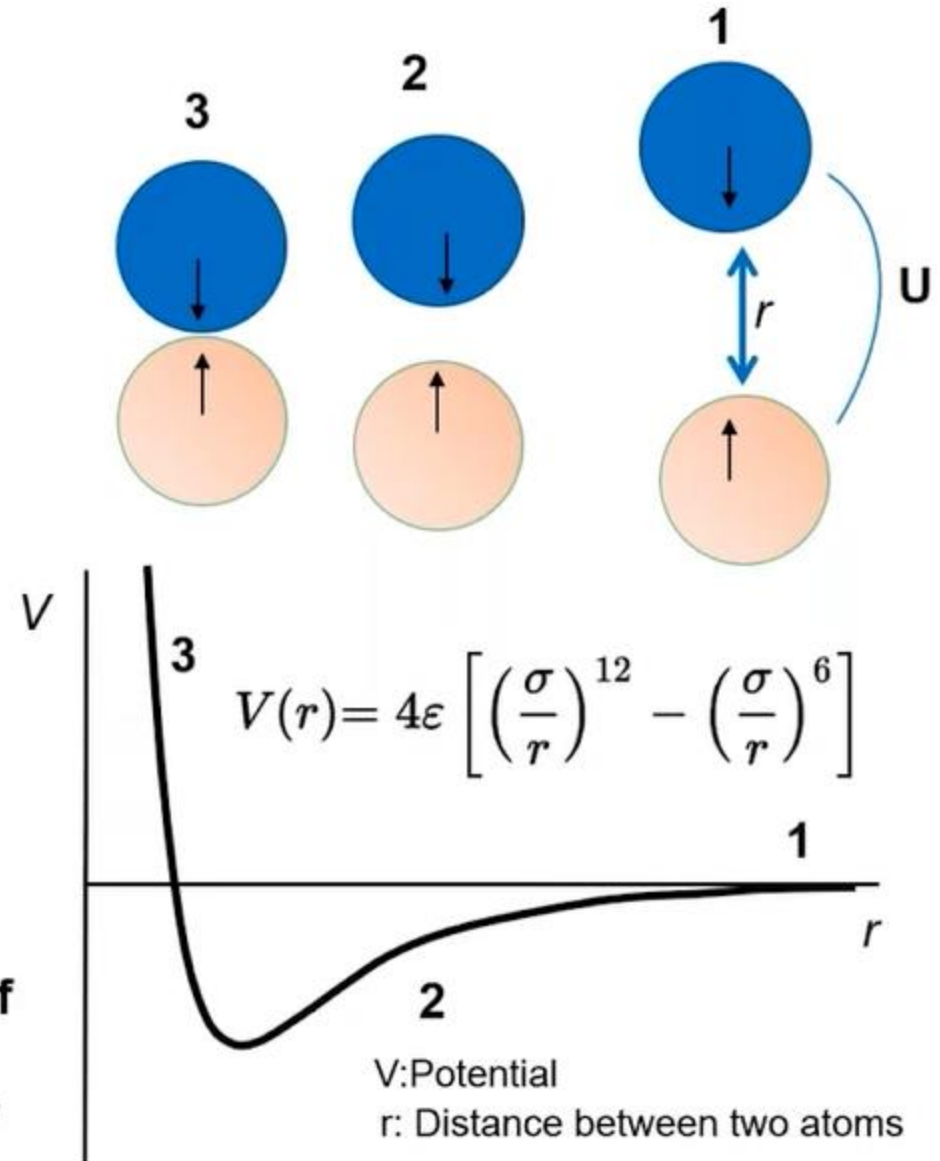
What are the forces on the AFM tip?



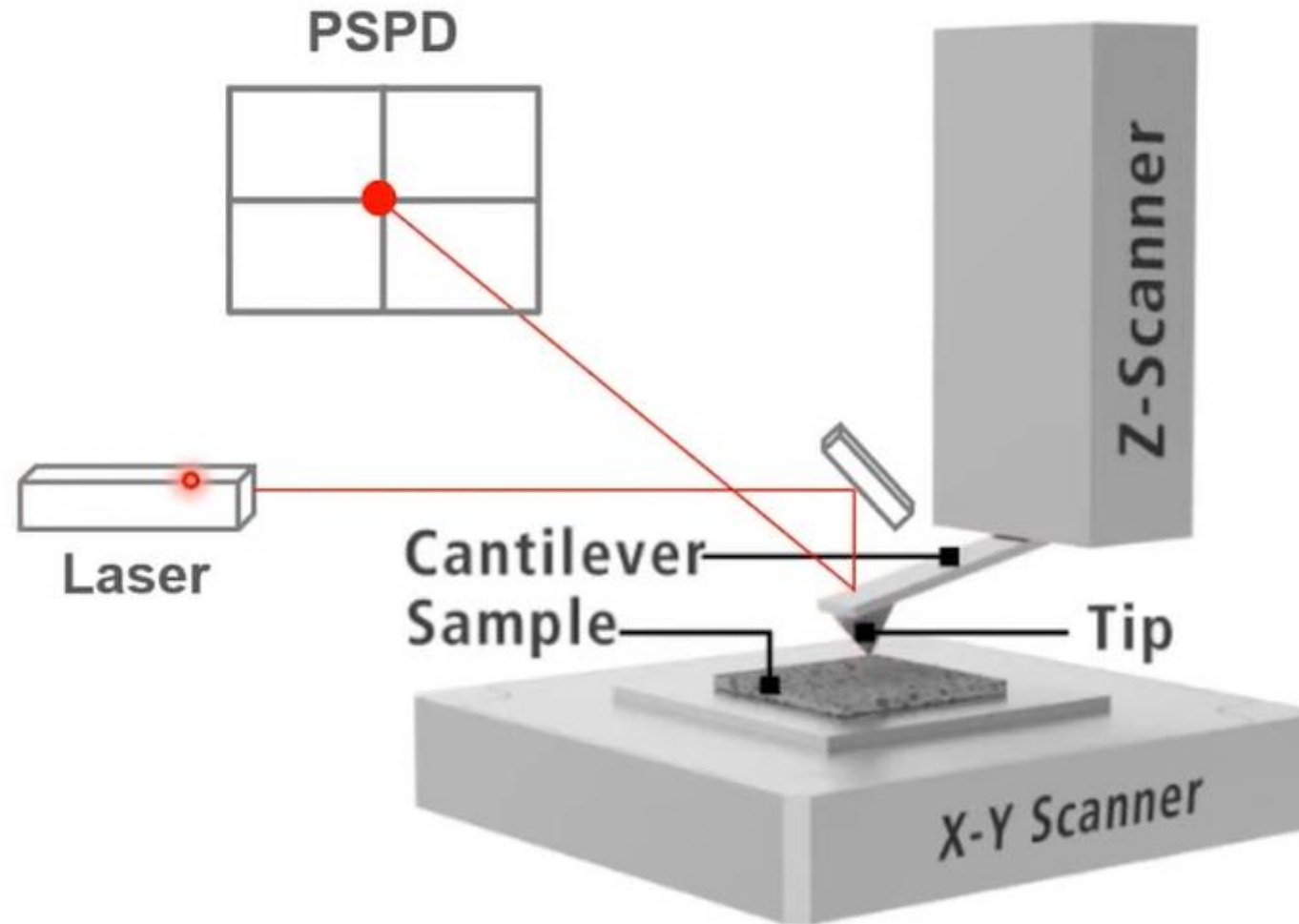
van der Waals forces



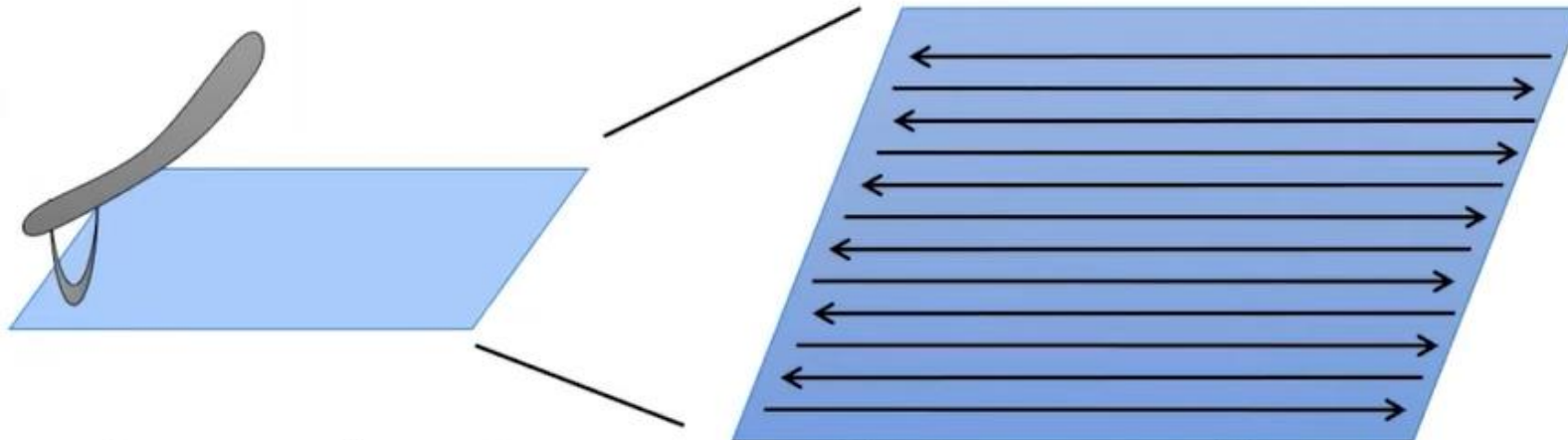
**Bending of
the
cantilever**



Key parts of AFM

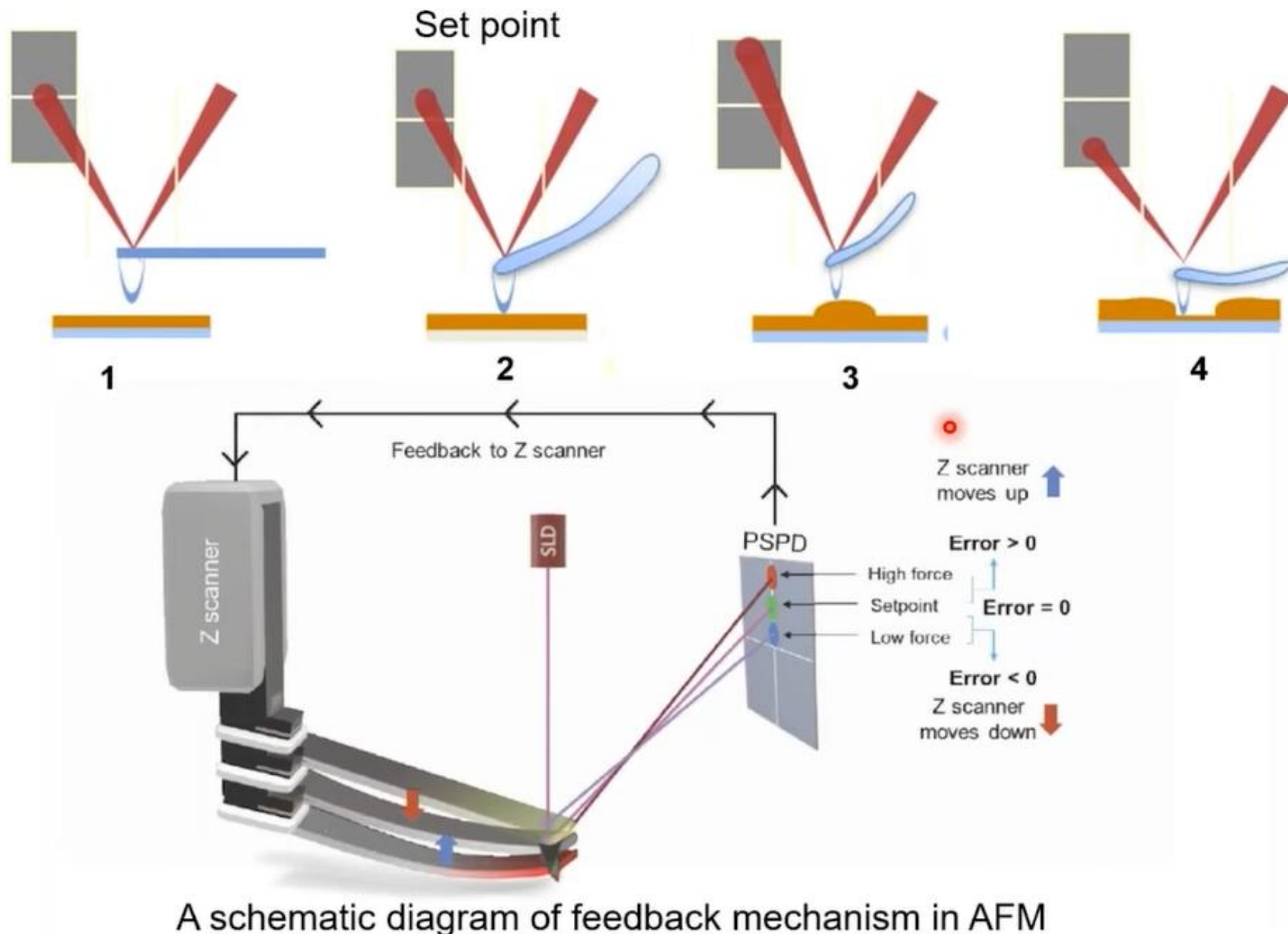


Imaging – Capture? or Scan?



Rectangular area of specimen

Raster scan pattern



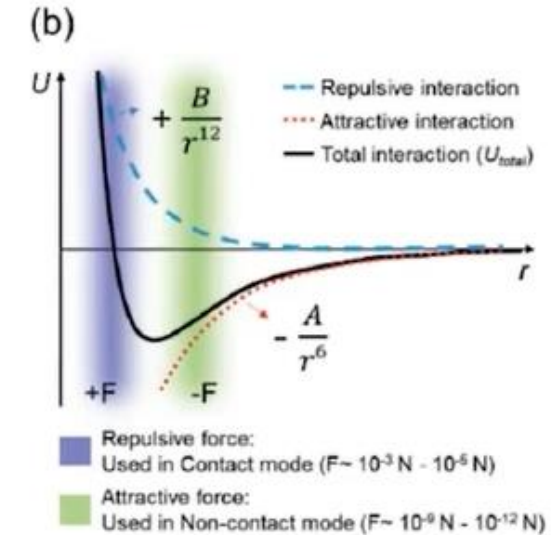
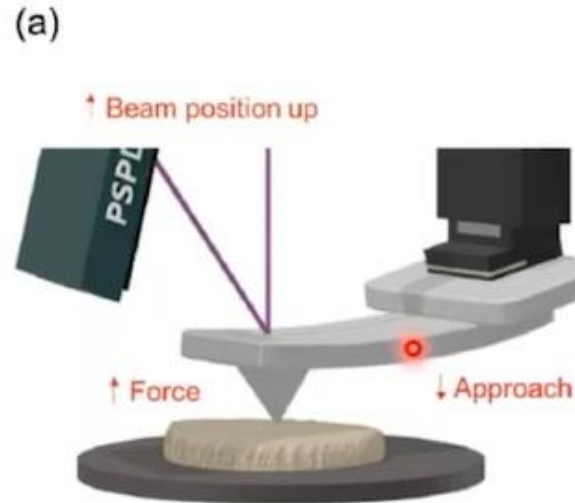
Imaging Modes

- **Contact mode**
- **Non-contact mode**
- **Tapping mode**

Contact mode

In this mode of operation of AFM the cantilever is in contact with the sample surface.

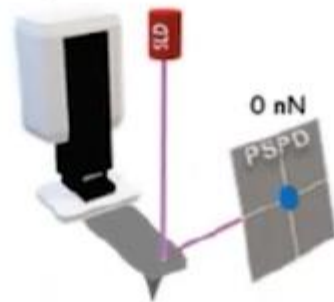
- **Setpoint** :- Specific value for interaction between tip and sample
- **Error signal**:- Setpoint - PSPD(A-B) signal
- **Feedback**:- Maintain constant Setpoint value by adjusting voltages applied to the Z Scanner



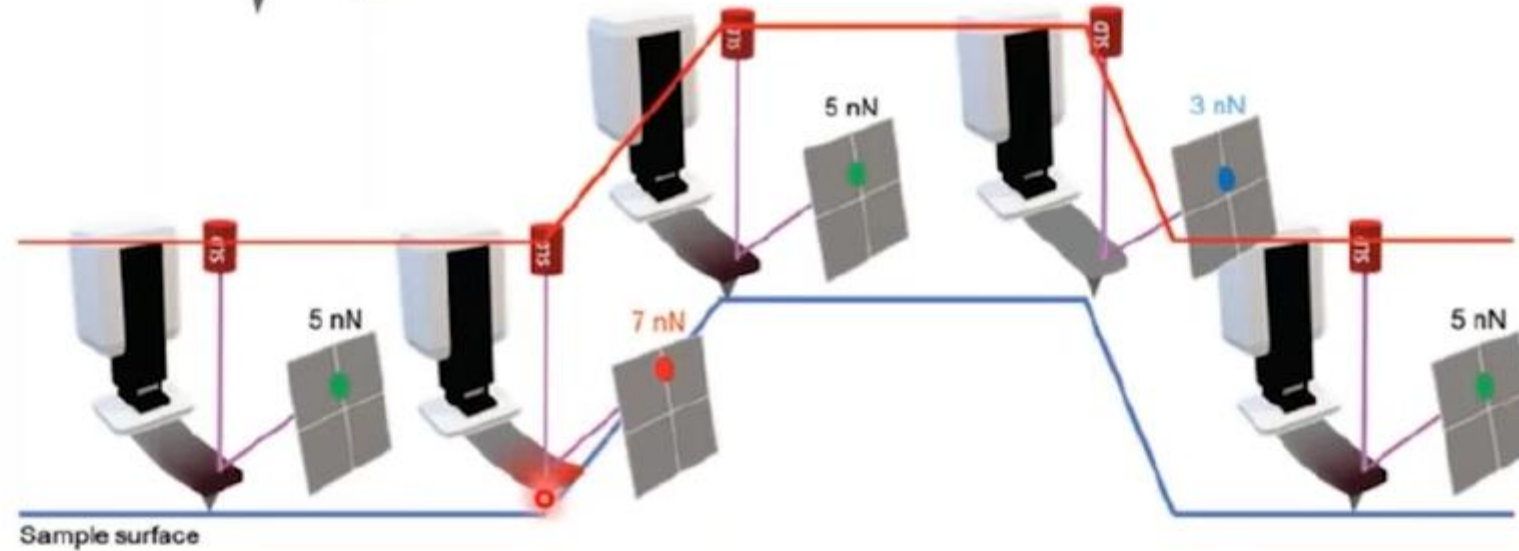
System tries to make zero value of error signal using z scanner movement

Contact mode

No tip-sample interaction before contact

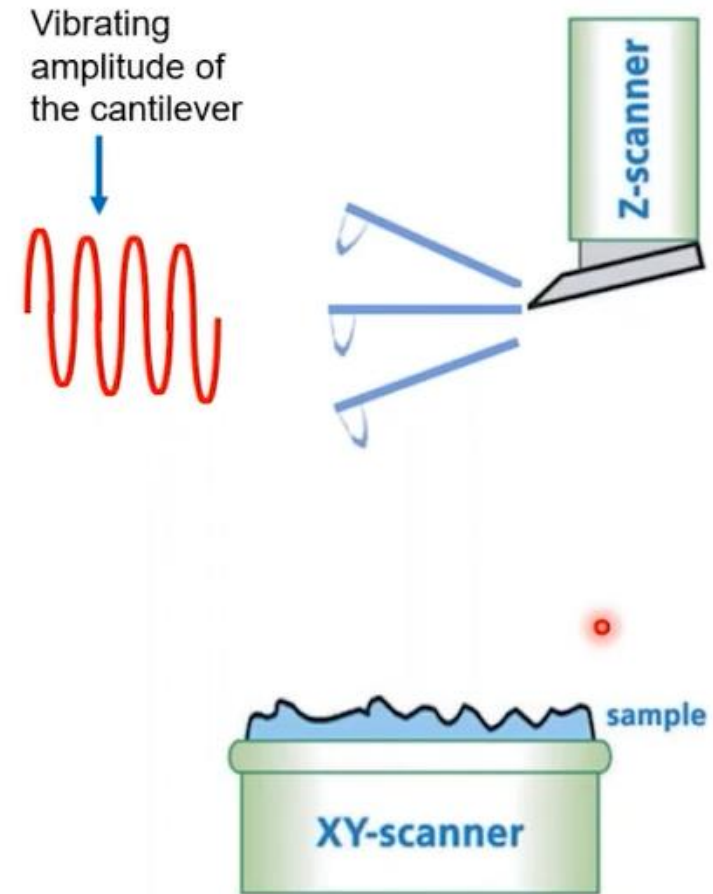
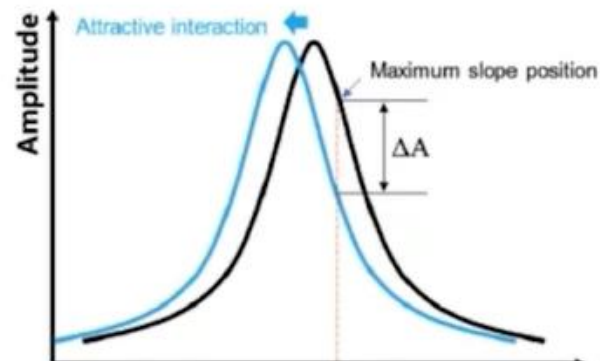
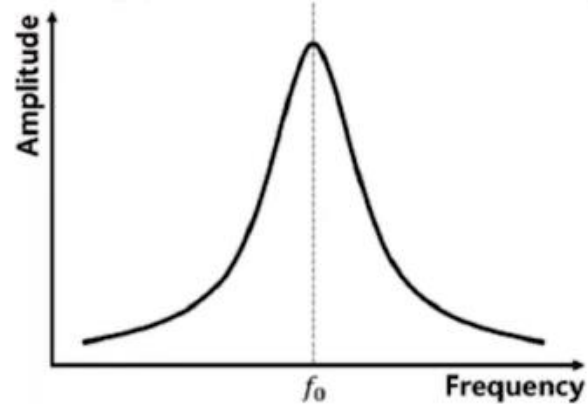


Sample height	Current force	Setpoint	Error	Z scanner
Flat	5 nN	5 nN	0	Hold
High	7 nN		+2	Up
Low	3 nN		-2	Down



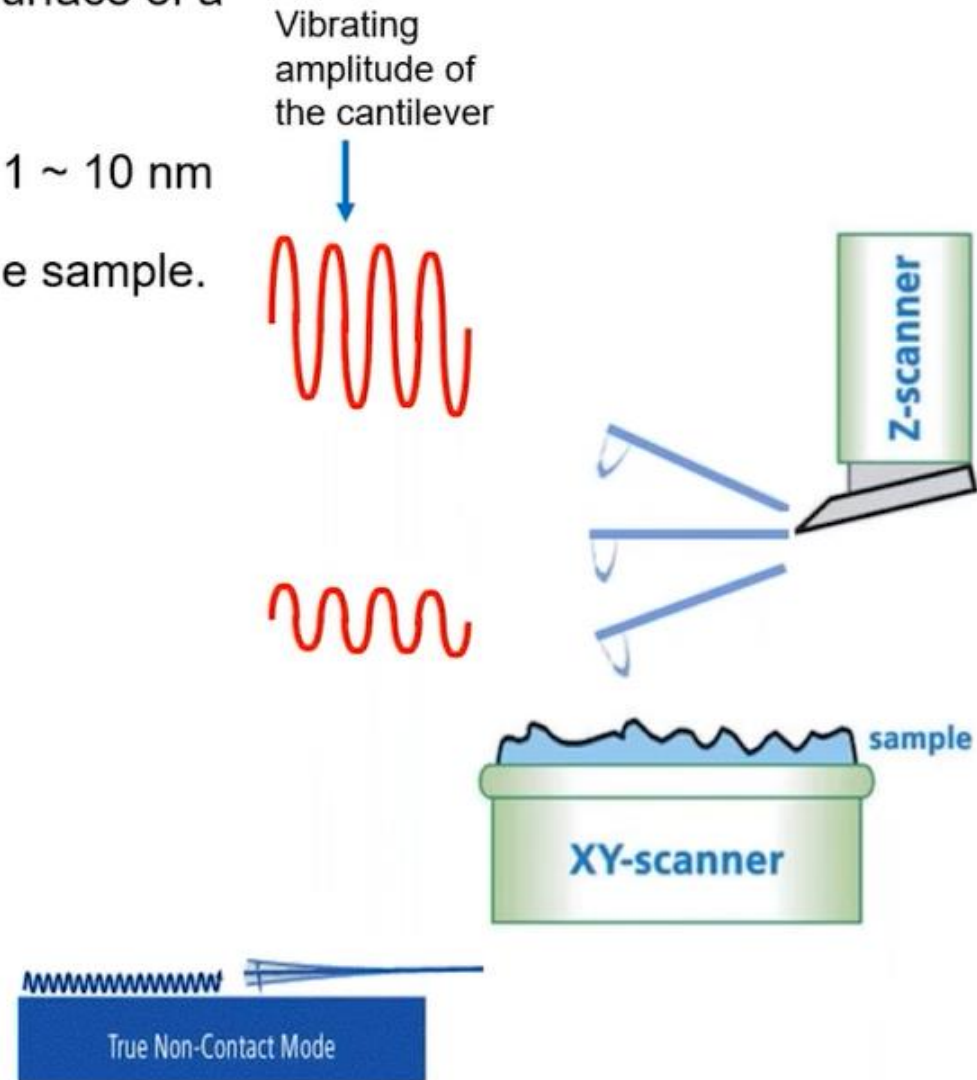
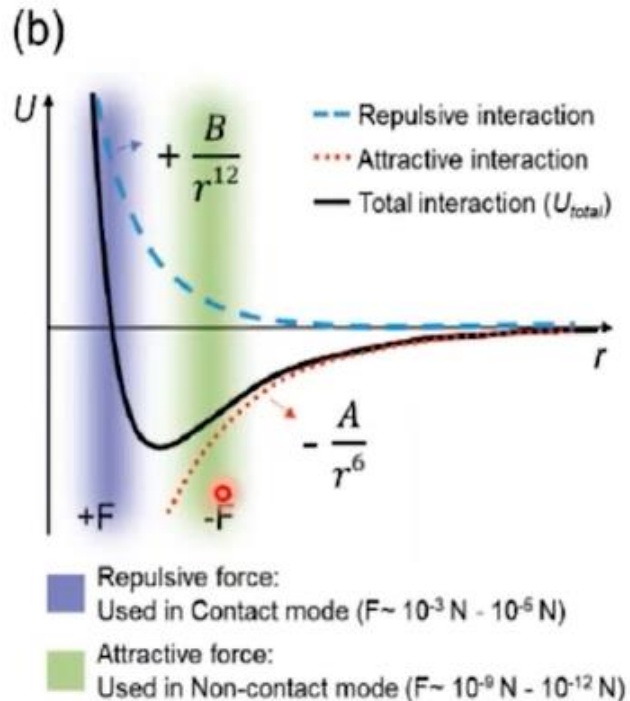
True Non-Contact™ Mode

- The cantilever is vibrated near the surface of a sample.
- Distance between tip and sample = 1 ~ 10 nm
- Vibrating probe does not damage the sample.

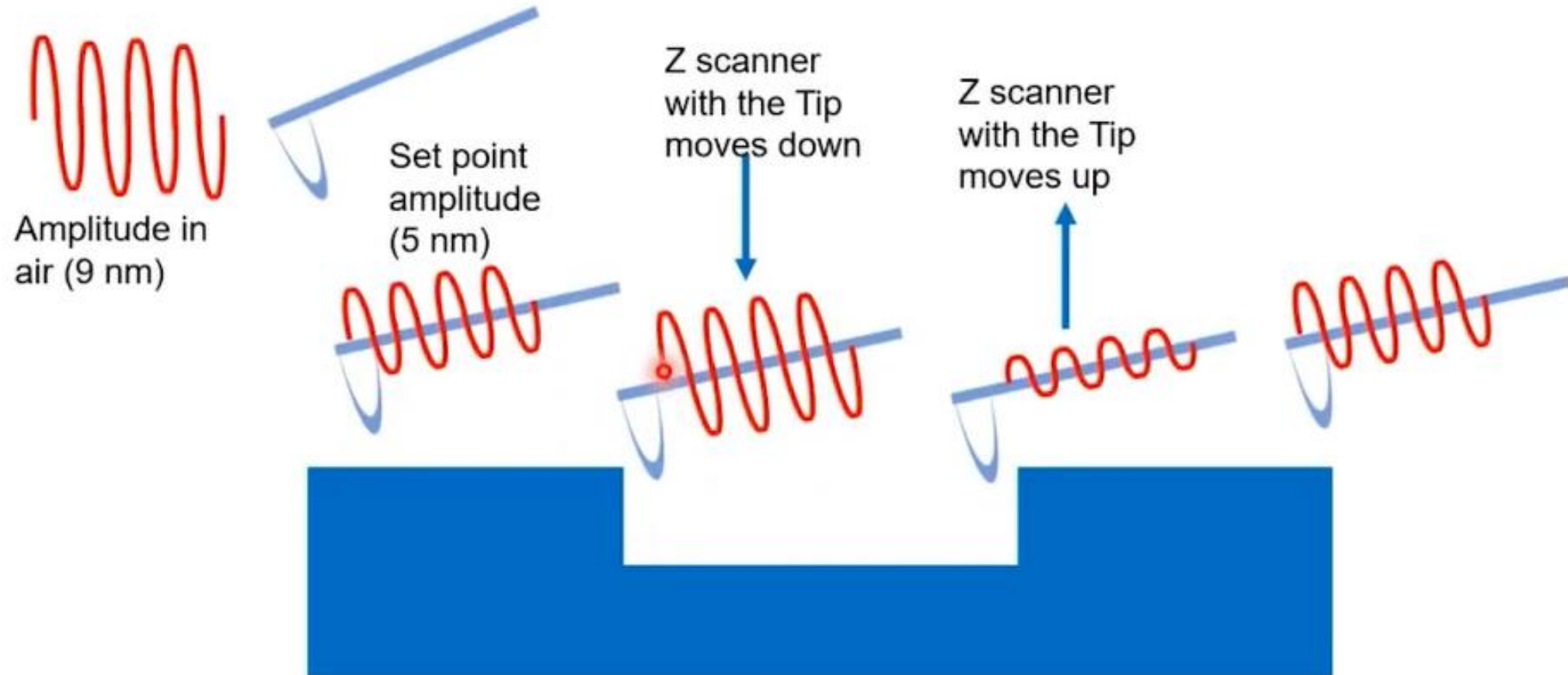


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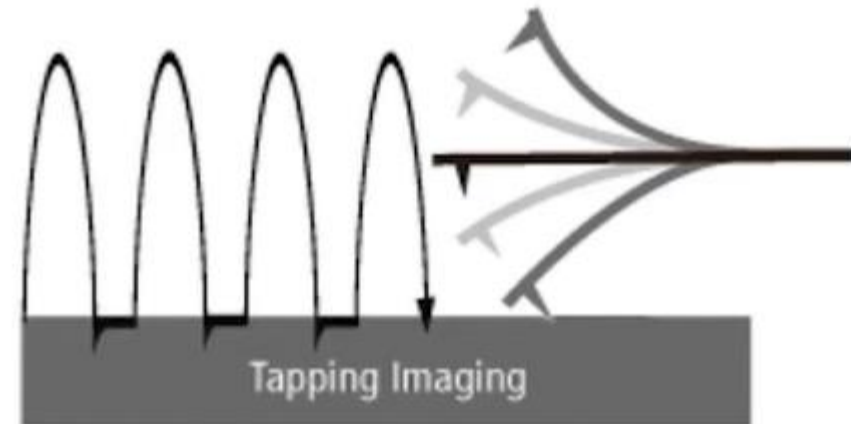


True Non-Contact™ Mode

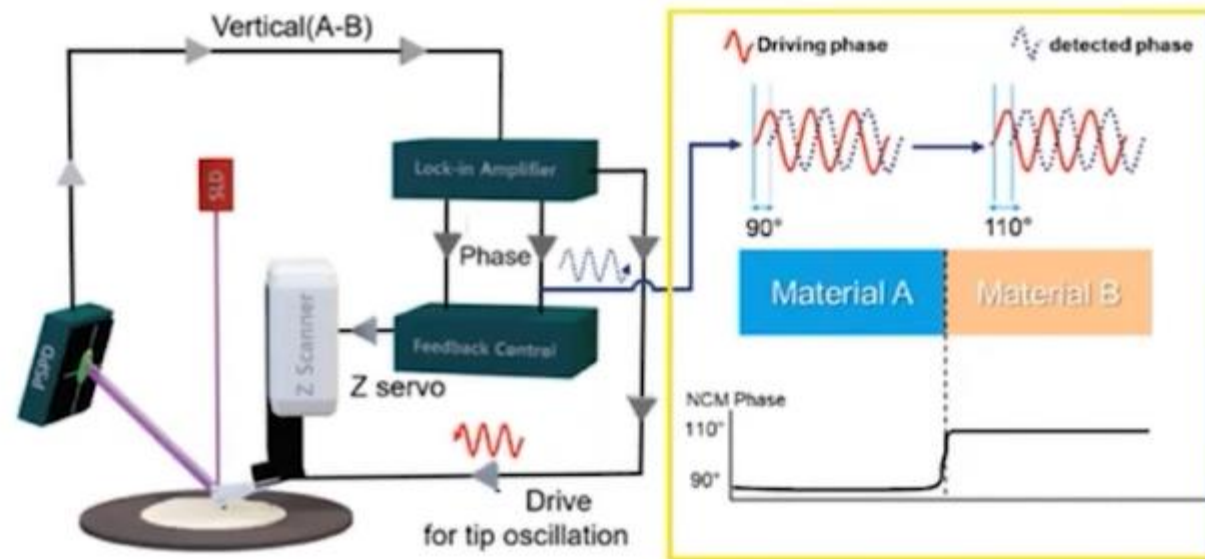


Intermittent or tapping mode

Tapping mode is a dynamic AFM technique that images the sample topography by scanning the surface with an oscillating cantilever, similar to Park's True Non-contact mode.



In Tapping mode the cantilever vibrates with larger amplitudes compared to True Non-contact mode and the tip makes intermittent contact with the sample surface during the measurement.



Tip Wearing (Accuracy & Repeatability)

- Tapping mode causes tip wearing
- Tip wearing constantly changes the tip geometry, making the measurement value non-repeatable.

